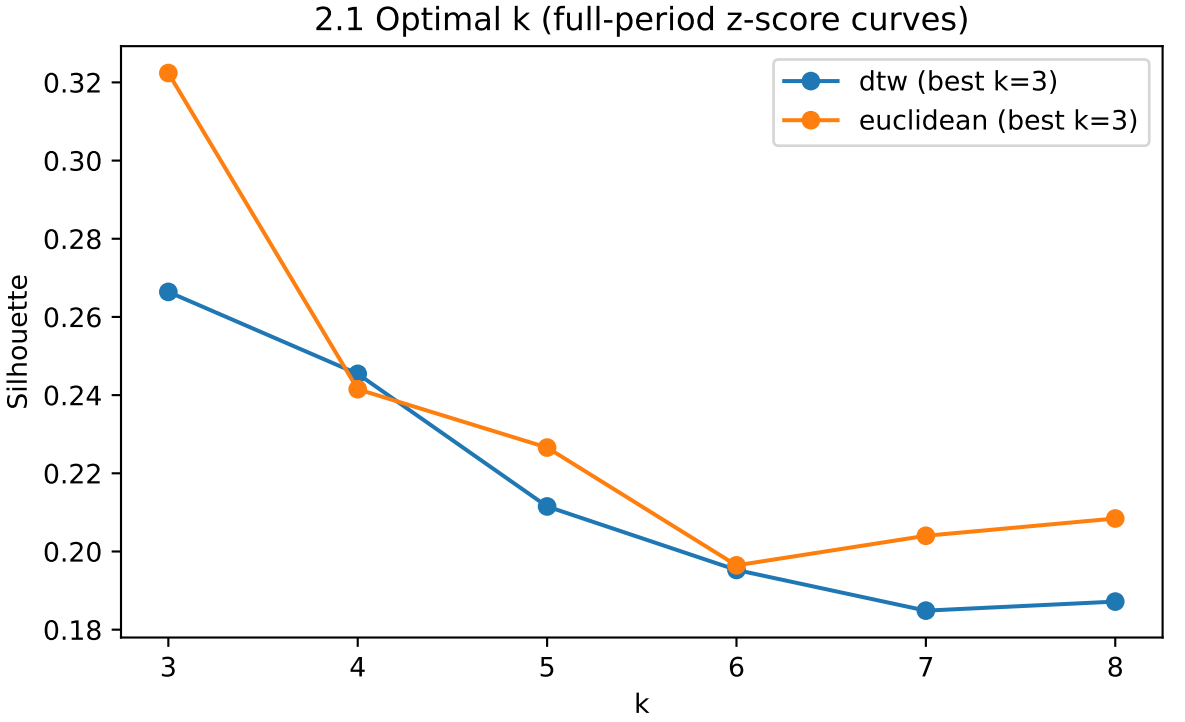
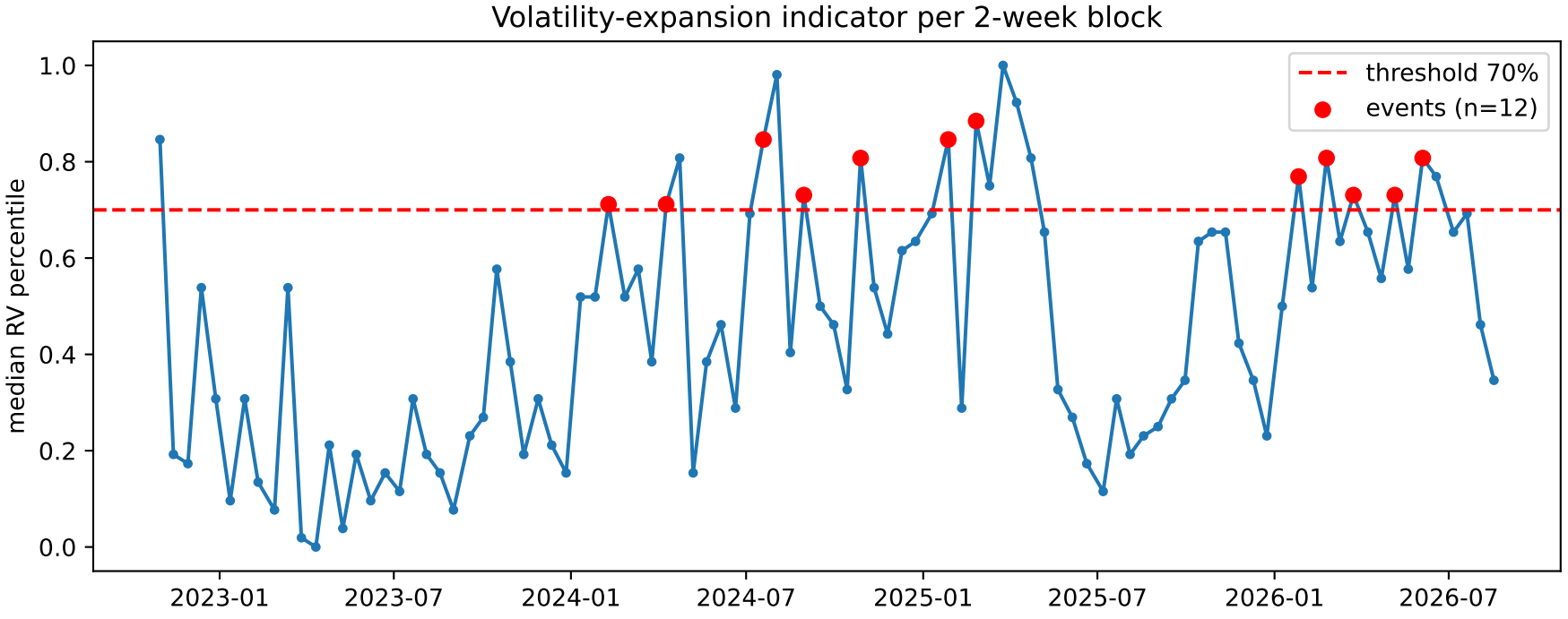
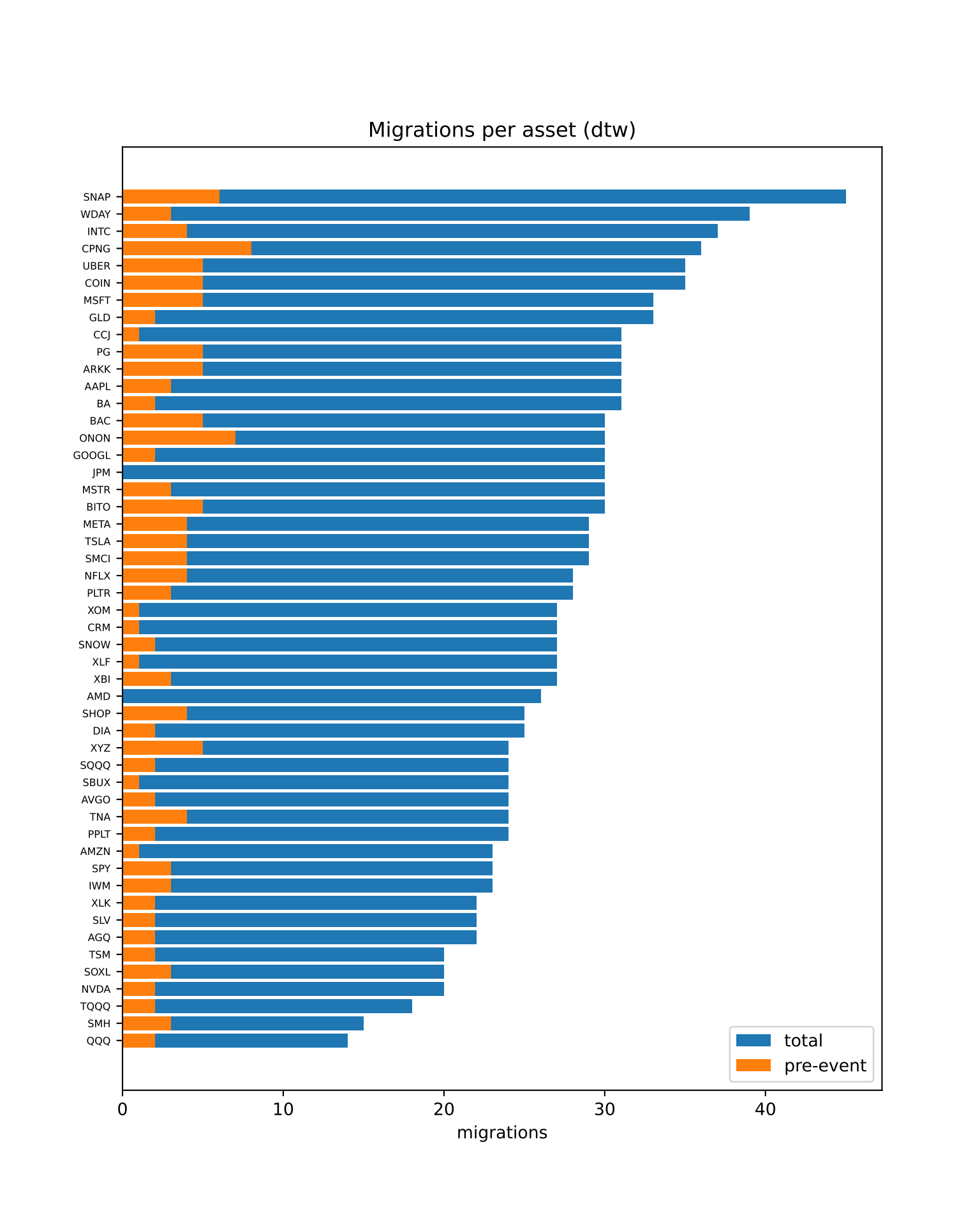
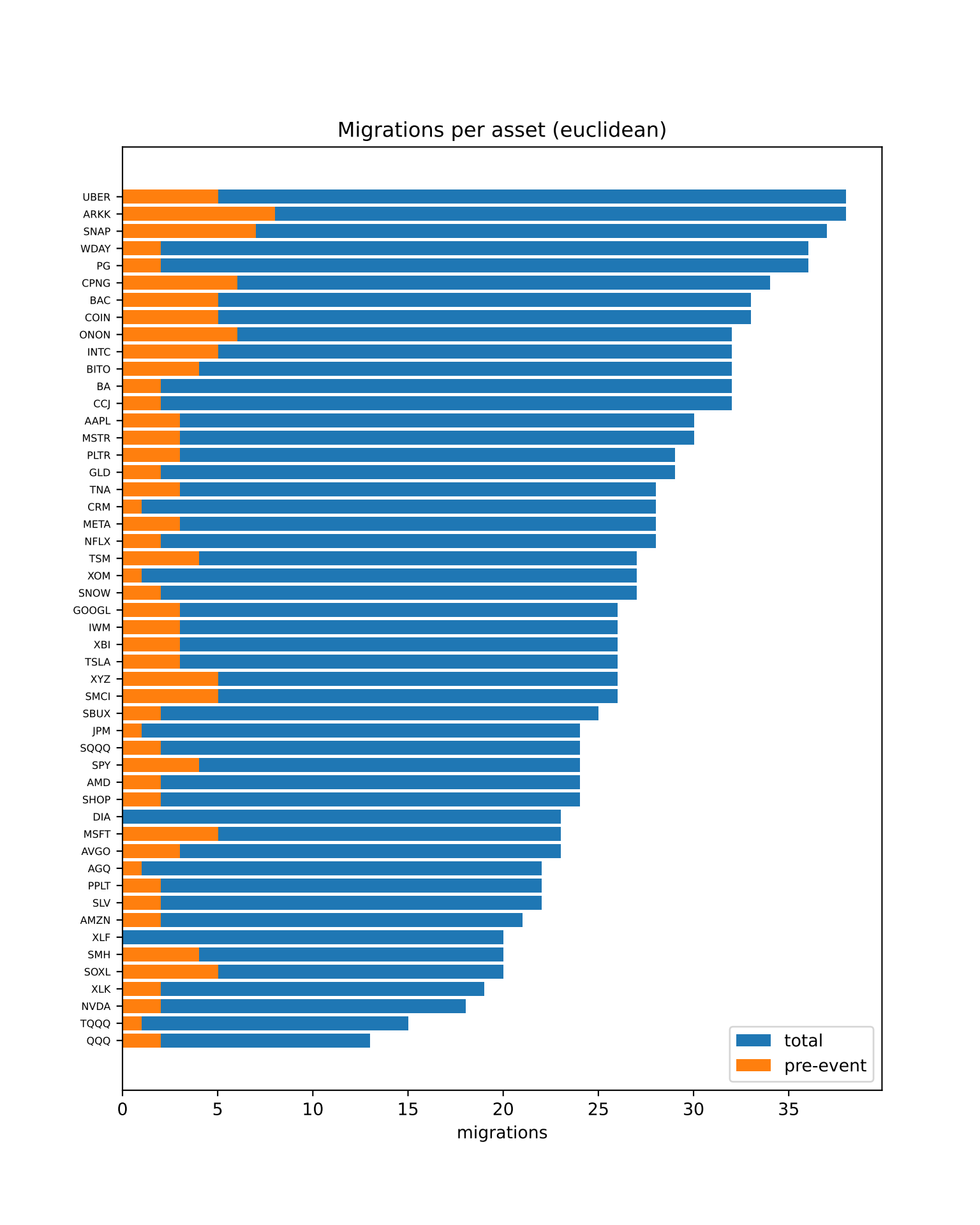

2.1 Optimal k (full-period z-score curves)
dtw (best k=3)
euclidean (best k=3)
Silhouette
0.32
0.30
0.28
0.26
0.24
0.22
0.20
0.18
3
4
5
6
7
8
k


Volatility-expansion indicator per 2-week block
median RV percentile
1.0
0.8
0.6
0.4
0.2
0.0
threshold 70%
events (n=12)
2023-01
2023-07
2024-01
2024-07
2025-01
2025-07
2026-01
2026-07


Migrations per asset (dtw)
SNAP
WDAY
INTC
CPNG
UBER
COIN
MSFT
GLD
CCJ
PG
ARKK
AAPL
BA
BAC
ONON
GOOGL
JPM
MSTR
BITO
META
TSLA
SMCI
NFLX
PLTR
XOM
CRM
SNOW
XLF
XBI
AMD
SHOP
DIA
XYZ
SQQQ
SBUX
AVGO
TNA
PPLT
AMZN
SPY
IWM
XLK
SLV
AGQ
TSM
SOXL
NVDA
TQQQ
SMH
QQQ
total
pre-event
0
10
20
30
40
migrations


Migrations per asset (euclidean)
UBER
ARKK
SNAP
WDAY
PG
CPNG
BAC
COIN
ONON
INTC
BITO
BA
CCJ
AAPL
MSTR
PLTR
GLD
TNA
CRM
META
NFLX
TSM
XOM
SNOW
GOOGL
IWM
XBI
TSLA
XYZ
SMCI
SBUX
JPM
SQQQ
SPY
AMD
SHOP
DIA
MSFT
AVGO
AGQ
PPLT
SLV
AMZN
XLF
SMH
SOXL
XLK
NVDA
TQQQ
QQQ
total
pre-event
0
5
10
15
20
25
30
35
migrations